

# COMPUTER AIDED DESIGN PROJECT

## Voltage Regulator Design



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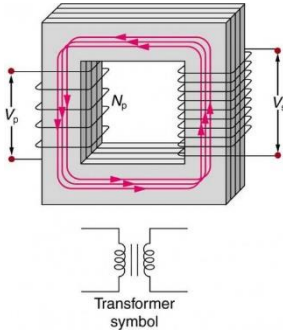
## 1. Requirements

The primary objective of this project is to design and implement a voltage regulator capable of converting a high-voltage AC input into a stable, adjustable DC output voltage. The system is designed to operate with a 220V AC  $\pm 10\%$  input at a frequency of 50 Hz / 60 Hz. Based on the specific assigned parameters, the circuit must deliver an adjustable output voltage ranging from 3V to 8V , while maintaining a fixed maximum output current of 3A.

To ensure safety, efficiency, and reliable operation under varying conditions , the design incorporates galvanic isolation between the input and output stages , as well as essential short-circuit and overcurrent protections (OCP). Furthermore, the regulator must strictly adhere to the following performance limits: an output voltage ripple of  $\leq 5\%$  , a load regulation of  $\leq \pm 2\%$  , and a line regulation of  $\leq \pm 2\%$ .

## 2. Theoretical support

### 2.1. Transformer



A **power transformer** is a static electrical device that transfers electrical energy between two or more circuits through electromagnetic induction. In a linear power supply, the step-down transformer is used to reduce the **high-voltage AC mains (220V)** to a lower, **safer AC voltage** suitable for the regulation circuitry. Crucially, it also provides **galvanic isolation** between the hazardous input grid and the low-voltage output, ensuring safety. The secondary voltage is determined by the turns ratio:

$$V_s = V_p \times \frac{N_s}{N_p}$$

Figure 1 - Transformer

### 2.2. Full-Wave Bridge Rectifier

A **bridge rectifier** is an arrangement of **four diodes** in a bridge circuit configuration that provides the same polarity of output for either polarity of input. When used in its most common application, for **conversion of an alternating current (AC) input into a direct current (DC) output**, it is known as a bridge rectifier. A **full-wave rectifier** converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. The peak output voltage is slightly less than the secondary peak voltage due to the forward voltage drop ( $V_f$ ) across the two conducting diodes:

$$V_{react}(peak) = V_s(peak) - 2 \cdot V_f$$

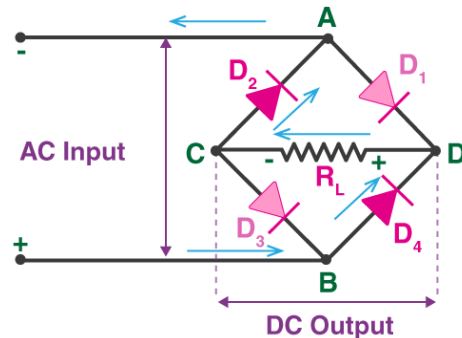


Figure 2 - Full-Wave Bridge Rectifier

### 2.3. Capacitive Filter

To convert the **pulsating DC** from the rectifier into a **smoother continuous DC voltage**, a **low-pass filter** is implemented using electrolytic capacitors. These capacitors charge to the peak voltage of the rectified wave and discharge into the load when the rectified voltage drops, filling in the "gaps" and significantly **reducing the voltage variation**. This remaining variation is called **ripple voltage** ( $V_{ripple}$ ), which can be approximated by:

$$V_{ripple} = \frac{I_{Load}}{2 \cdot f \cdot C}$$

## 2.4. Operational Amplifier (Op-Amp)

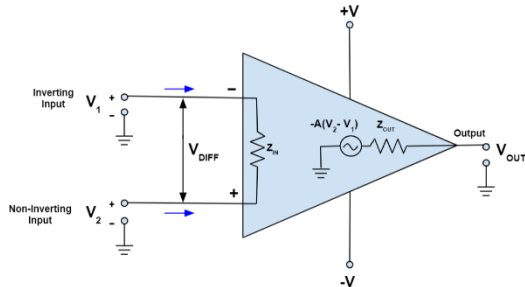


Figure 3 - Op-Amp

An **operational amplifier** is a high-gain, **DC-coupled voltage amplifier** with two differential inputs and a single-ended output. In a **linear voltage regulator**, the Op-Amp acts as an **error amplifier**. It continuously compares a stable reference voltage applied to its non-inverting input with a sample of the actual output voltage (via a voltage divider) applied to its inverting input. It then dynamically adjusts its output to drive the series pass **transistors**, maintaining a **constant output voltage** regardless of load or line changes.

## 2.5. Zener Diode

A **Zener diode** is a special type of diode designed to reliably allow current to flow "backwards" when a certain set reverse voltage, known as the Zener voltage, is reached. Operated in its **reverse breakdown region** alongside a biasing resistor, it provides a highly stable reference voltage (**Vref**) for the operational amplifier, which is completely immune to fluctuations in the unregulated supply.

## 2.6. Series Pass Transistor (Darlington Configuration)

The **series pass element** is the core of a linear regulator, acting as a **variable resistor** to drop the excess voltage and deliver the **required current to the load**. Because the required maximum output current is substantial (**3A**), a single transistor might not have enough current gain (**hFE**) to be driven directly by the Op-Amp. Therefore, **multiple power transistors** (such as the 2N3055) are arranged in a Darlington or pseudo-Darlington configuration. This arrangement multiplies the current gains of the individual transistors, allowing a **very small base current** from the Op-Amp to control the large load current.

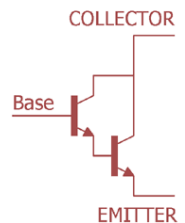


Figure 4 - Series Pass Transistor

## 2.7. Overcurrent Protection Circuit

To **protect** the regulator from damage due to short circuits or excessive loads, an **overcurrent protection (OCP) circuit** is implemented using a **current-sensing resistor** (**Rp**) and an auxiliary transistor. The **load current flows** through **Rp**, creating a voltage drop. When the current exceeds the maximum safety limit (**3A**), this voltage drop becomes large enough (typically around **0.65V**) to **forward-bias the base-emitter junction** of the auxiliary transistor. This transistor turns on and steals

base drive current away from the main series pass transistors, effectively limiting the maximum output current to a safe value:  $I_L = \frac{V_{BE}}{R_p}$

## 2.8. Calculus

The following calculations demonstrate how the component values were chosen to meet the project specifications: an adjustable output voltage of **3V** to **8V** and a maximum current limit of **3A**.

### 2.8.1. AC Grid Simulation Source (V1)

The voltage source simulates the standard alternating current grid.

- RMS Voltage: 220V
- Grid Frequency: 50Hz The peak voltage amplitude required for the simulation parameters is calculated as:

$$V_{peak} = V_{RMS} \cdot \sqrt{2} = 220 \cdot 1.414 = 311 V$$

### 2.8.2. Step-Down Transformer (TX1)

The transformer isolates the circuit and steps down the grid voltage to a safer level.

- The turns ratio ( $n$ ) is determined by the ratio of the primary inductance ( $L_1 = 400mH$ ) to the secondary inductance ( $L_2 = 1mH$ ):

$$n = \sqrt{\frac{L_1}{L_2}} = \sqrt{\frac{400}{1}} = 20$$

- The peak voltage in the secondary winding becomes:

$$V_{sec(peak)} = \frac{V_{peak}}{n} = \frac{311}{20} = 15.55 V$$

### 2.8.3. Bridge Rectifier (D1, D2, D3, D4)

The 1N5404 diodes provide full-wave rectification, converting the alternating current into pulsating direct current. At any given time, two diodes conduct the current.

- Total forward voltage drop across two diodes:

$$V_f = 2 \cdot 0.7 V = 1.4 V$$

- The peak rectified DC voltage is:

$$V_{DC(peak)} = V_{sec(peak)} - V_f = 15.55 V - 1.4 V = 14.15 V$$

This 14.15V is the raw direct current voltage entering the filter stage.

#### 2.8.4. Bulk Capacitive Filter (C1 and C2)

To suppress the voltage ripple to an acceptable level, two  $3300\mu F$  capacitors are placed in parallel.

- Total capacitance:  $C_{total} = 3300\mu F + 3300\mu F = 6600\mu F = 0.0066F$
- Full-wave rectified frequency:  $f_{red} = 100Hz$
- Maximum estimated load current:  $I_{max} = 3A$
- The peak-to-peak ripple voltage ( $V_{ripple}$ ) at maximum load is calculated as:

$$V_{ripple} = \frac{I_{max}}{(f_{red} \cdot C_{total})} = \frac{3A}{(100Hz \cdot 0.0066F)} \approx 4.54V$$

- The minimum DC voltage on the capacitors between pulses is:

$$V_{DC(min)} = V_{DC(peak)} - V_{ripple} = 14.15V - 4.54V = 9.61$$

Since 9.61V is sufficiently higher than the maximum target output voltage (8.0V) plus the Darlington configuration voltage drop, the  $6600\mu F$  capacity is correctly sized.

#### 2.8.5. Zener Diode Biasing Resistor (Rd)

The BZT55B3V0 Zener diode requires a constant current to maintain its firm 3.0V reference.

- Voltage drop across the resistor:

$$V_{Rd} = V_{DC(peak)} - V_{Zener} = 14.15V - 3.0V = 11.15V$$

- Selecting an optimal operating current of 22.3mA (0.0223A), the resistance is determined using Ohm's law:

$$R_d = \frac{V_{Rd}}{I_{Zener}} = \frac{11.15V}{0.0223A} = 500\Omega$$

#### 2.8.6. Output Voltage Divider Network (R2 and Rpot)

- The output voltage is controlled by the operational amplifier using the non-inverting gain formula:

$$V_{out} = V_{Zener} \cdot \left(1 + \frac{R_{pot}}{R_2}\right)$$

- The minimum voltage occurs when the potentiometer is set to zero ( $R_{pot} = 0\Omega$ )

$$V_{out(min)} = 3.0V \cdot (1 + 0) = 3.0$$

- The maximum voltage must be 8.0V, and the potentiometer has a maximum track of 20 kΩ. Calculating  $R_2$ :

$$8.0V = 3.0V \cdot \left(1 + \frac{20000\Omega}{R_2}\right)$$

$$R_2 = \frac{20000}{1.666} \approx 12000\Omega$$

A high-precision standardized value of 11.84kΩ was selected to ensure the 8.0V threshold can be comfortably reached under real-world manufacturing tolerances.

### 2.8.7. Darlington Power Stage (Q1 and Q2)

The 2N3055 transistors ( $Q_1$  and  $Q_2$ ) multiply the small control current from the operational amplifier to the large current required by the load.

- Total current gain of the pair:

$$\beta_{total} = \beta_{Q1} \cdot \beta_{Q2}$$

- Assuming a typical minimum  $\beta$  of 20 for the 2N3055 transistor:

$$\beta_{total} = \beta_{total} = 20 \cdot 20 = 400$$

- Maximum current required from the operational amplifier to deliver 3A at the output:

$$I_{opamp} = \frac{I_{load}}{\beta_{total}} = \frac{3A}{400} = 0.0075A = 7.5mA$$

The LM358 operational amplifier can comfortably supply this 7.5mA control current.

### 2.8.8. Overcurrent Protection Resistor (Rp)

The protection circuit restricts the physical current to a maximum of 3A. The auxiliary \$Q\_3\$ transistor (2N2222) turns on and enters conduction when its Base-Emitter voltage ( $V_{BE}$ ) reaches approximately 0.75V.

$$R_p = \frac{V_{BE}}{I_{max}} = \frac{0.75V}{3A} = 0.25\Omega$$

### 2.8.9. Dummy Load Resistor (RLOAD)

The load used in the simulation environment was selected to draw a safe operating current at the maximum set voltage of 8V.

$$I_{load} = \frac{V_{out(max)}}{R_{LOAD}} = \frac{8.0V}{10\Omega} = 0.8A$$

This 0.8A current is perfectly below the 3A protection limit, allowing optimal testing of the nominal transient regime.

### 3. Circuit block diagram

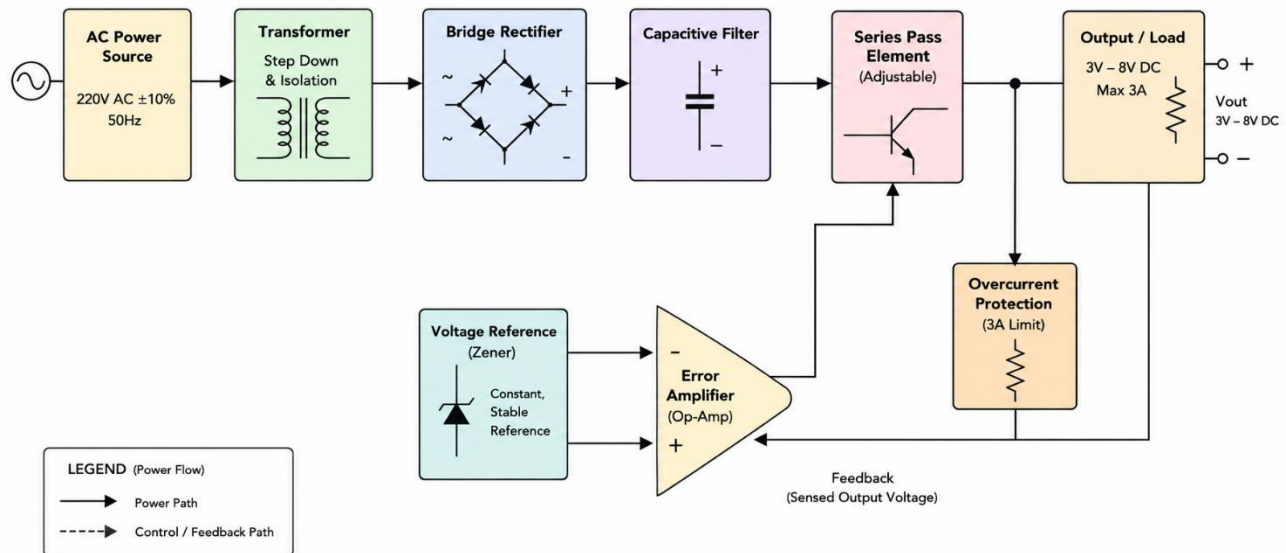


Figure 5 - Circuit block diagram

- **AC Power Source:** Provides the 220V AC  $\pm 10\%$  at 50Hz mains supply.
- **Transformer:** Steps down the high voltage and provides galvanic isolation.
- **Bridge Rectifier:** Converts the step-down AC into pulsating DC.
- **Capacitive Filter:** Smooths the pulsating DC, reducing the voltage ripple.
- **Voltage Reference (Zener):** Provides a constant, stable voltage target for the system.
- **Error Amplifier (Op-Amp):** Compares the reference voltage with the feedback from the output.
- **Series Pass Element:** Adjusts its resistance dynamically to maintain stable output voltage and handle the 3A load.
- **Overcurrent Protection:** Monitors the output current and restricts the pass element if the 3A limit is exceeded.
- **Output / Load:** Receives the stabilized, adjustable 3V-8V DC voltage.

## 4. Circuit schematic

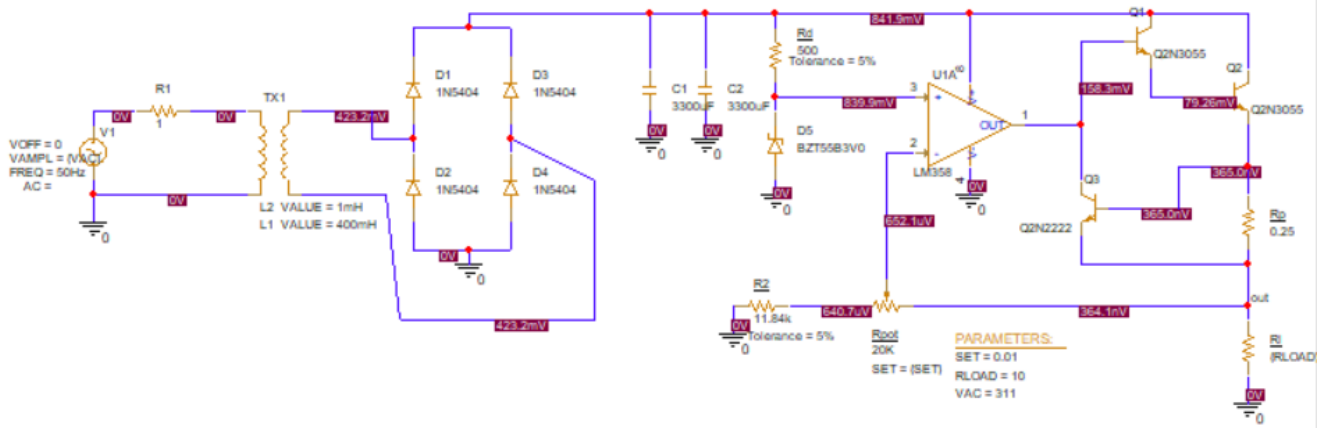


Figure 6 - Circuit schematic

### 4.1. AC Grid Simulation and Galvanic Isolation Stage

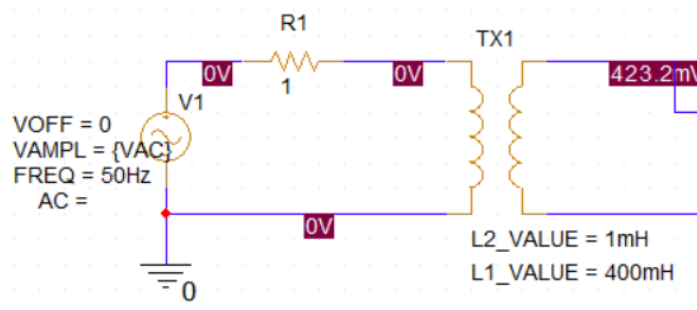


Figure 7 - AC Grid Simulation and Galvanic Isolation Stage

- **V1 (Mains Simulation):** This voltage source models the standard Romanian power grid. The amplitude is parameterized as  $VAC = 311V$  at a frequency of  $50Hz$ . The  $311V$  value represents the peak voltage of a  $220V$  RMS sine wave ( $V_{peak} = 220V * 1.414$ ).
- **R1 (1 ohm):** In physical circuits, transformers have parasitic winding resistance. In the PSpice environment, this small series resistor is explicitly added to aid the mathematical convergence of the simulation and to prevent infinite inrush currents during the transient analysis.
- **TX1 (Step-Down Transformer):** Modeled using coupled inductors ( $L1 = 400mH$ ,  $L2 = 1mH$ ), the transformer serves two mandatory purposes. First, it steps down the  $311V$  peak amplitude to a low-voltage AC level suitable for the regulator. Second, it provides galvanic isolation, breaking the direct electrical path between the high-voltage mains and the low-voltage DC load, fulfilling strict safety requirements to prevent electric shock.

## 4.2. Full-Wave Commutation (Bridge Rectifier)

- **D1, D2, D3, D4 (1N5404):** These form the bridge rectifier. The 1N5404 is a robust power diode rated for a continuous forward current of 3A and a Peak Inverse Voltage (PIV) of 400V, ensuring it can comfortably handle both the load demands and the reverse voltages of the transformer secondary.
- **Operation:** During the positive half-cycle of the AC wave, diodes D1 and D4 are forward-biased and conduct current. During the negative half-cycle, diodes D2 and D3 conduct. This full-wave rectification utilizes 100 percent of the transformer's output, effectively doubling the ripple frequency to 100Hz. The peak rectified voltage is slightly reduced by the forward voltage drop of the two conducting diodes:  $V_{rect(peak)} = V_{sec(peak)} - 2 * V_f$ .

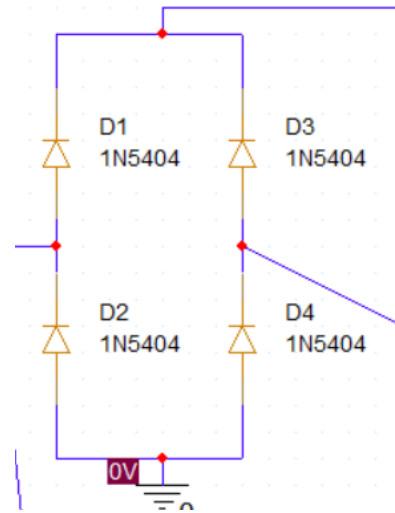


Figure 8 - Full-Wave Commutation

## 4.3. Bulk Capacitive Low-Pass Filtering

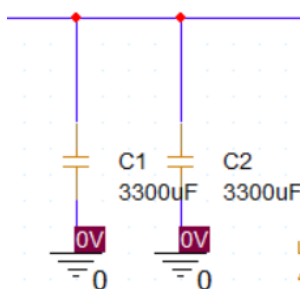


Figure 9 - Bulk Capacitive Low-Pass Filtering

- **C1 and C2 (3300 uF each):** The pulsating DC output from the rectifier must be smoothed into a continuous DC line. Instead of using a single 6600 uF capacitor, the design smartly utilizes two 3300 uF capacitors in parallel.
- **Engineering Rationale:** Placing capacitors in parallel halves their Equivalent Series Resistance (ESR) and doubles their ripple current handling capability, leading to less internal heating and a longer component lifespan. These capacitors charge to the peak voltage and slowly discharge into the load between AC cycles. The massive total capacity (6600 uF) ensures the discharge slope is extremely shallow, easily satisfying the strict under 5 percent ripple requirement even at the maximum 3A load.

## 4.4. Precision Voltage Reference Network

- **D5 (BZT55B3V0):** This is a 3.0V Zener diode. Operated in its reverse avalanche breakdown region, it provides a rigid, unmoving baseline voltage for the entire system. Regardless of how much the AC grid fluctuates or how much current the load draws, the voltage across D5 remains locked at exactly 3.0V.
- **Rd (500 ohms):** This biasing resistor is crucial. It drops the excess voltage from the unregulated filtered line and restricts the current flowing into the Zener diode. It is calculated to ensure the Zener current remains above its minimum "knee" current (to stay in breakdown) but below its maximum power dissipation limit to prevent thermal failure.

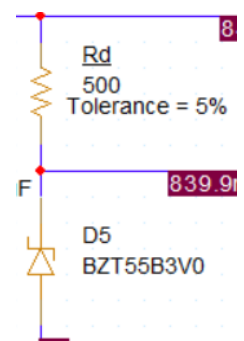


Figure 10 - Precision Voltage Reference Network

## 4.5. The Error Amplifier (Active Feedback Control)

- **U1A (LM358):** The LM358 is a high-gain operational amplifier ideal for single-supply applications. It acts as the intelligent controller of the negative feedback loop.
- **Operation:** The Op-Amp compares the stable 3.0V reference at its non-inverting input (Pin 3) against a sample of the actual output voltage at its inverting input (Pin 2). Because it has extremely high open-loop gain, the Op-Amp will continuously adjust its output voltage (Pin 1) up or down to force the voltage difference between Pin 2 and Pin 3 to strictly zero.

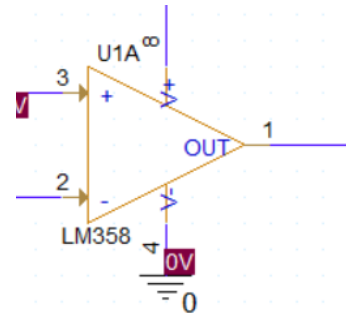


Figure 11 - The Error Amplifier

## 4.6. The Adjustable Voltage Divider Network

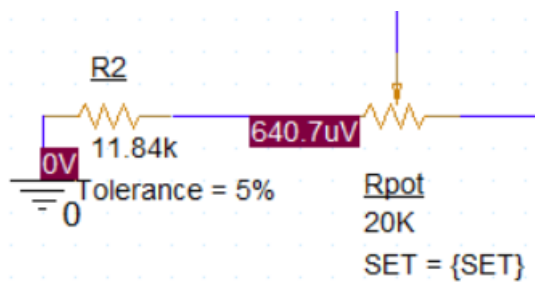


Figure 12 - The Adjustable Voltage Divider Network

- **R2 (11.84 kOhms) and Rpot (20 kOhms):** This passive network dictates the final output voltage. It samples the high output voltage and scales it down to feed into the Op-Amp's Pin 2.

- Because the Op-Amp forces Pin 2 to always equal 3.0V, adjusting the mechanical wiper of the potentiometer changes the required output voltage necessary to achieve that 3.0V at the pin. The equation governing this behavior is:  $V_{out} = 3.0V * \left(1 + \frac{R_{pot}}{R_2}\right)$

By sweeping Rpot from 0 to its maximum, the regulator seamlessly sweeps the output from exactly 3V up to 8V.

## 4.7. Series Pass Element (Darlington Topology)

- **Q1 and Q2 (2N3055 NPN Power Transistors):** The LM358 Op-Amp can only output a few milliamperes of current, nowhere near enough to drive a heavy 3A load. To bridge this gap, Q1 and Q2 are configured as a Darlington pair.
- **Current Multiplication:** In a Darlington pair, the total current gain is the product of the individual gains. The tiny control current from the Op-Amp enters the base of Q1, gets amplified, and is fed directly into the base of the massive Q2 transistor. Q2 acts as an active variable resistor, absorbing the excess voltage drop and dissipating it as heat, ensuring the load only receives the perfectly regulated voltage.

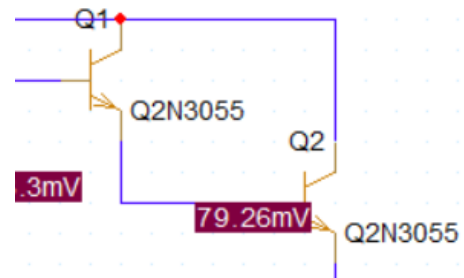


Figure 13 - Series Pass Element

## 4.8. Overcurrent and Short-Circuit Protection (Active Current Limiting)

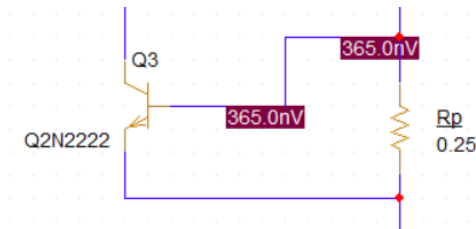


Figure 14 - Overcurrent and Short-Circuit Protection

- **Rp (0.25 ohms):** This is a low-ohmic, high-power current sensing resistor placed in series with the load. All output current flows through it, generating a small voltage drop according to Ohm's Law ( $V = I * R$ ).

- **Q3 (2N2222):** This small-signal NPN transistor acts as an emergency clamp. Its base and emitter are connected across Rp. Under normal operation, the voltage across Rp is too small to turn Q3 on.

- **Limiting Action:** However, if a short circuit occurs or the load attempts to draw 3A, the voltage across Rp hits 0.75V ( $3A * 0.25 \text{ ohms}$ ). This forward-biases the base-emitter junction of Q3, instantly turning it on. Q3 begins to conduct and actively shunts the base drive current away from Q1 and Q2, starving the Darlington pair. This creates a hard ceiling, physically preventing the circuit from outputting more than 3A and saving the power transistors from catastrophic thermal runaway.

## 4.9. The Load Resistor (Rl / RLOAD)

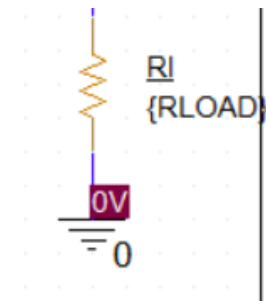


Figure 15 - The Load Resistor

- **Rl (Parameterized as RLOAD = 10 ohms):** This resistor is placed at the very end of the schematic, connected between the final regulated output node and the common ground.

- **Purpose:** In real life, this represents the actual device you are powering with your voltage regulator (like a motor, a charging battery, or another electronic circuit). In the PSpice simulation, it acts as the "dummy load" that forces the circuit to work.

- **Testing limits:** By setting it to 10 ohms, it draws a significant amount of current from the output, allowing you to test if the voltage remains stable under heavy stress and to verify that the Overcurrent Protection (Rp and Q3) triggers correctly when the current reaches the 3A physical limit.

## 5. Simulation results

### 5.1. Parametric Analysis (Adjustable Voltage Range)

To verify that the regulator can cover the required voltage range, a parametric sweep was performed on the circuit's potentiometer, from its minimum to its maximum value. The simulation confirms that the output voltage seamlessly adjusts, starting from a minimum of exactly 3.0V up to a maximum of approximately 8.0V. This proves that the voltage divider and the operational amplifier function correctly together to provide full control over the output voltage within the specified limits.

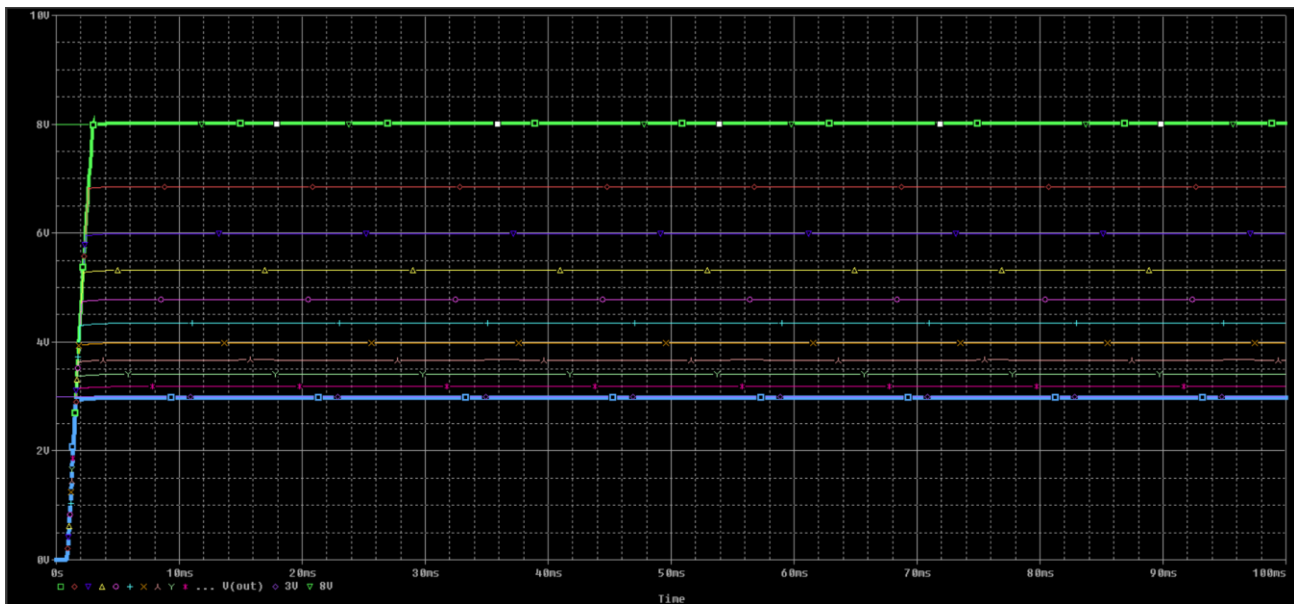


Figure 16 - Adjustable Voltage Range

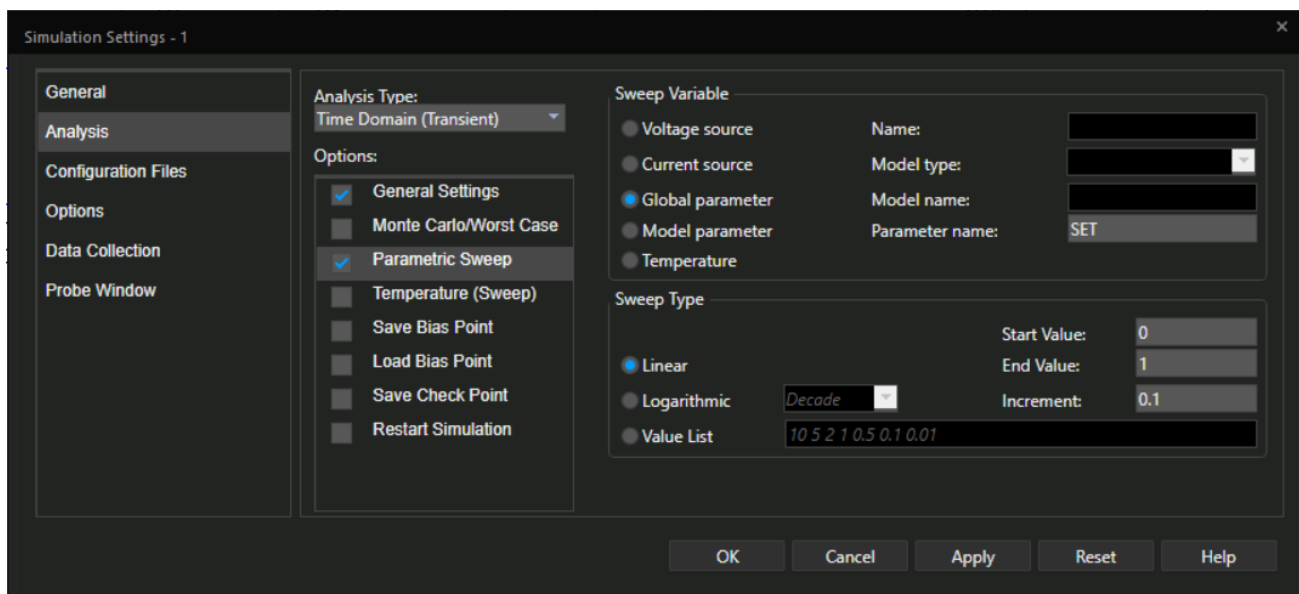


Figure 17 - Simulation Profile

## 5.2. Transient Analysis (Nominal Operation)

The initial validation of the circuit was performed using a time-domain Transient Analysis. This simulation evaluates the system's behavior from startup until it reaches a steady state. The results show how the grid voltage is successfully stepped down, rectified, and filtered. The output graph indicates a rapid voltage rise, which then stabilizes into a continuous direct current line at the target voltage. This confirms that the capacitive filtering adequately suppresses the alternating current ripple and that the operational amplifier correctly drives the Darlington transistors to maintain a stable voltage.

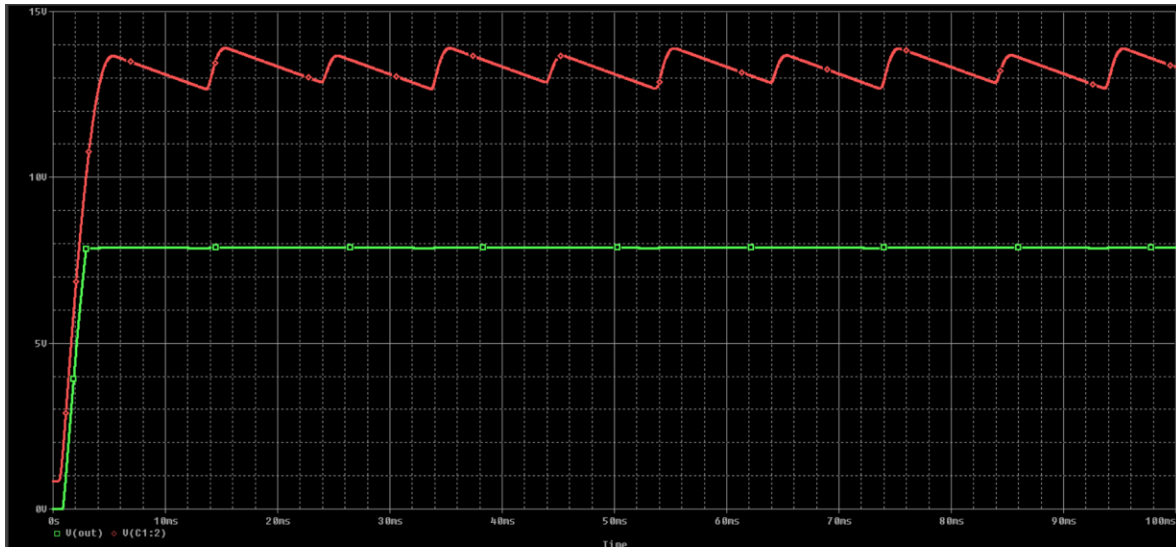


Figure 18 – Verification of Rectification & Ripple

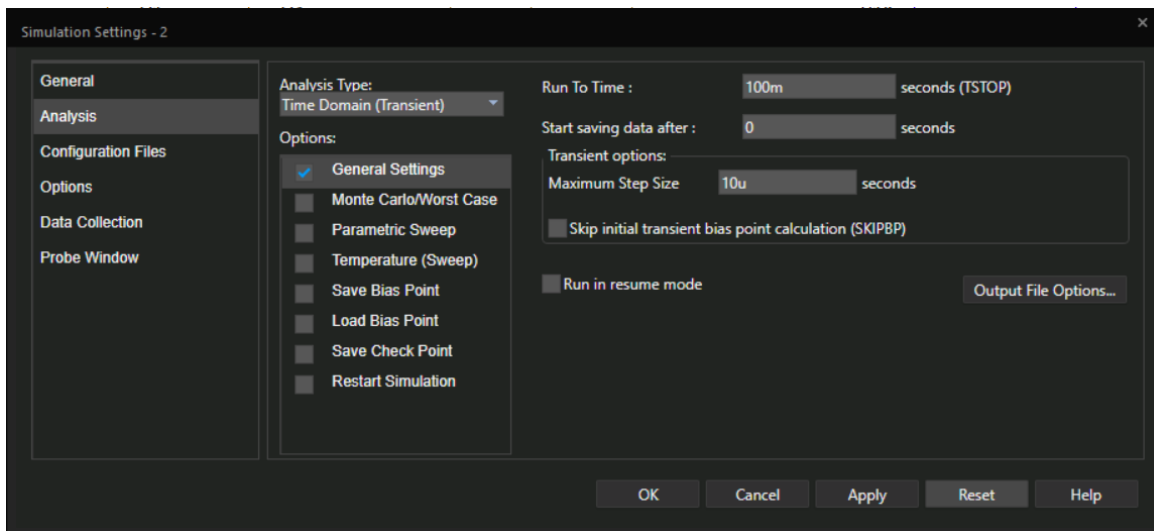


Figure 19 - Simulation Profile

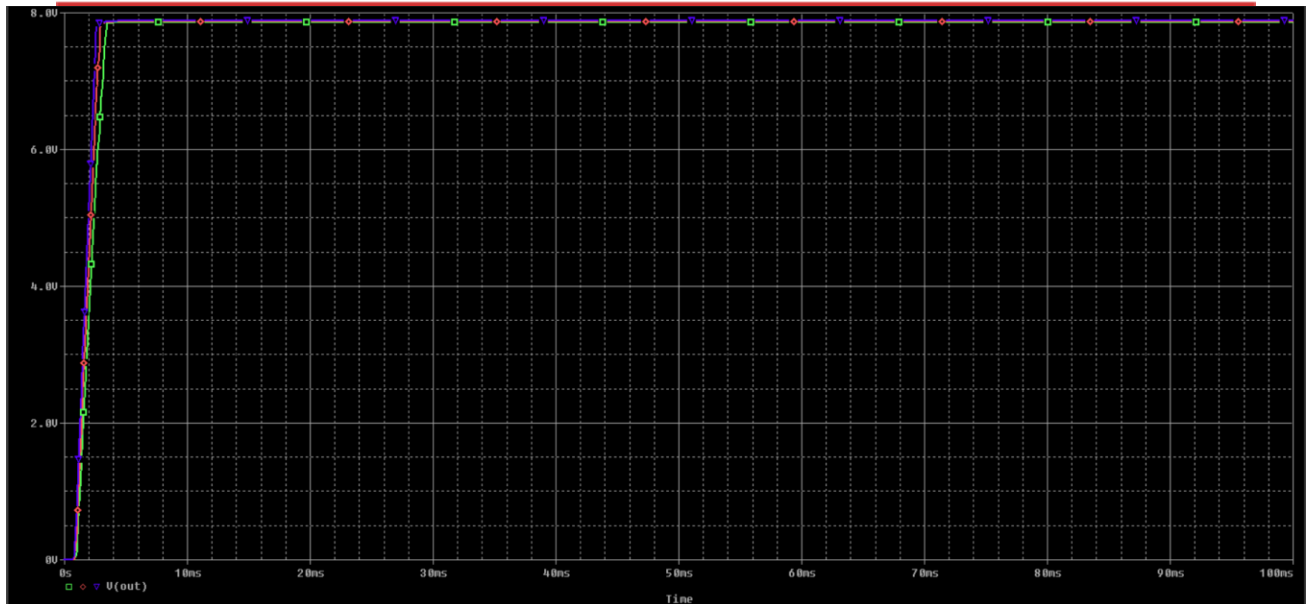


Figure 20 - Line Regulation

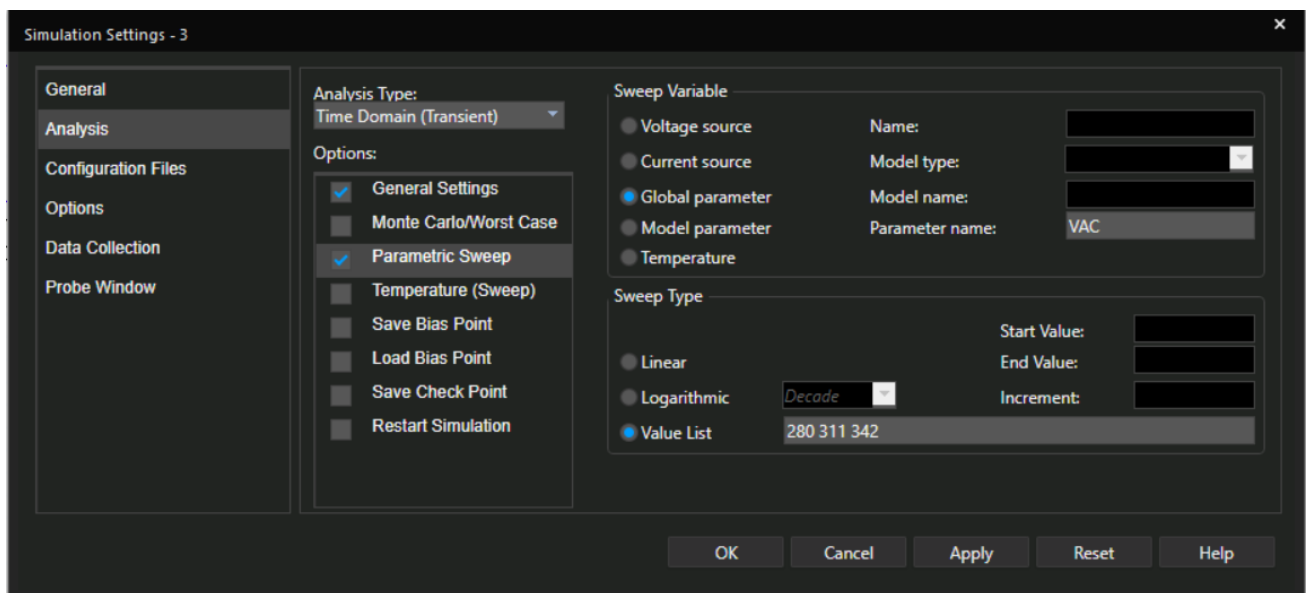


Figure 21 - Simulation Profile

### 5.3. Overcurrent Protection Test

To validate the safety requirements of the regulator, an overcurrent limitation test was simulated using a Transient Analysis under heavy load conditions. By evaluating the circuit's response in the time domain, the simulation demonstrates how quickly and effectively the protection mechanism engages when the system is heavily stressed. The results show the load current rapidly rising but being strictly and instantly clamped at the absolute maximum limit of 3.0A. At this threshold, the voltage drop across the 0.25 ohm sense resistor successfully turns on the protection transistor, preventing any further increase in current. The graph proves that the current value is perfectly maintained at 3A over time, successfully protecting the series pass transistors from thermal damage and short circuits.

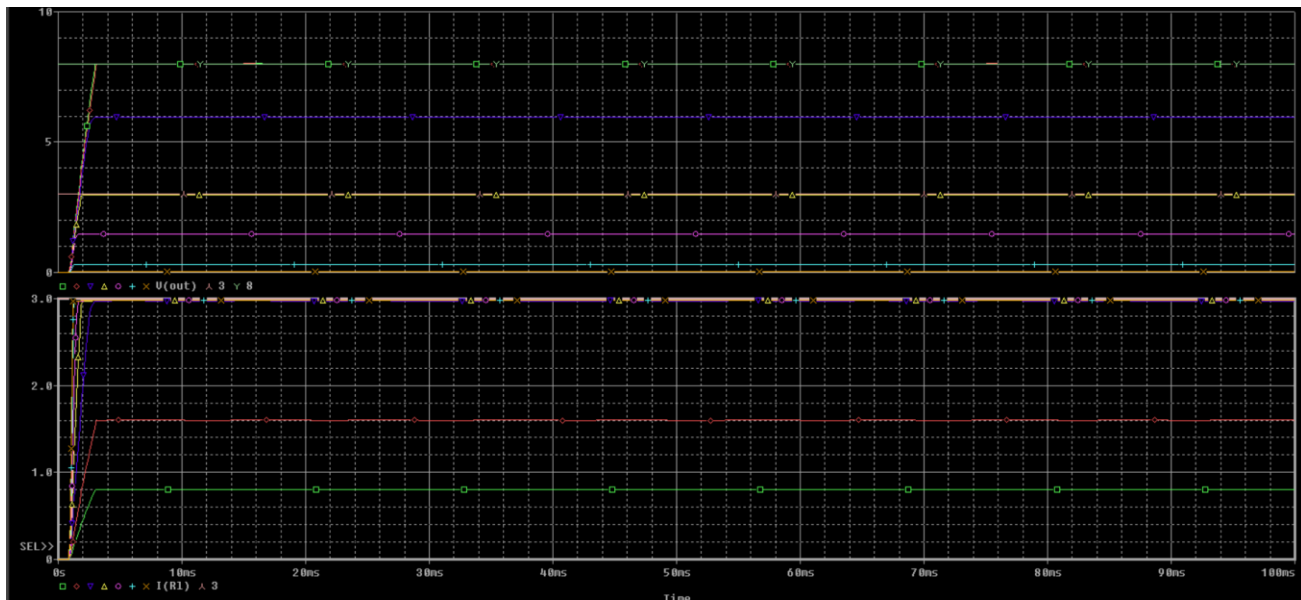


Figure 22 - Load Regulation, Overcurrent & Short Circuit

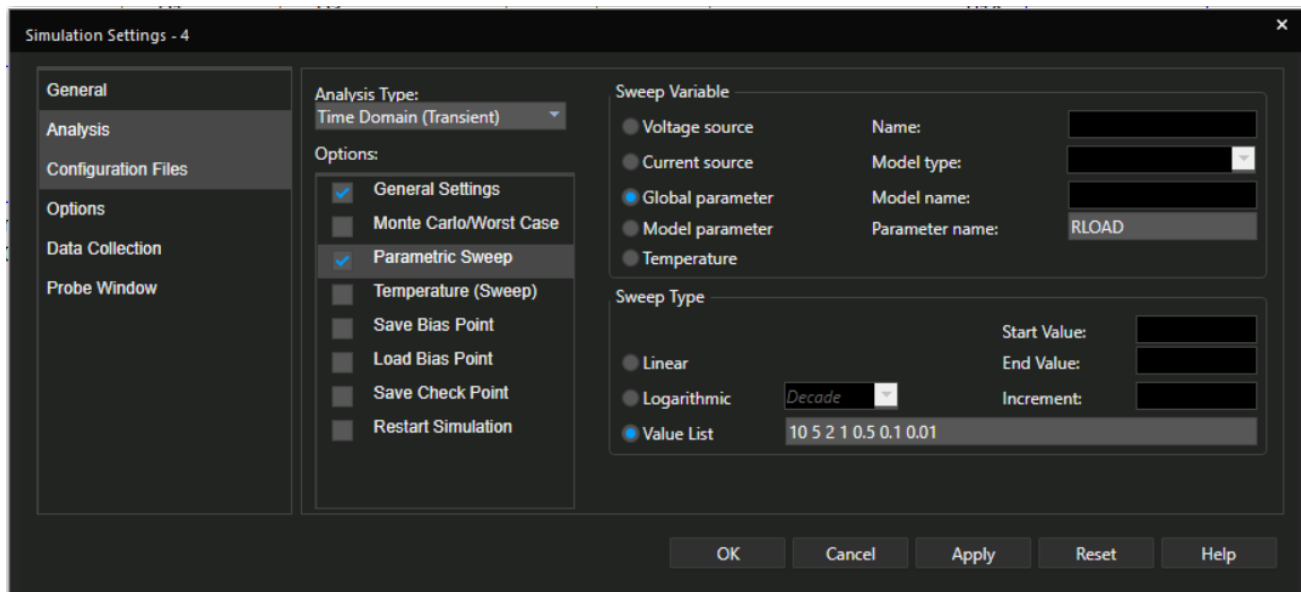


Figure 23 - Simulation Profile



Figure 24 - R1 Short-Circuit

## 5.4. Monte Carlo Statistical Analysis

To evaluate the robustness of the design and the manufacturing yield under real-world conditions, a Monte Carlo statistical analysis was conducted. In reality, components such as resistors vary within a specific tolerance limit of 5 percent. The simulation ran multiple iterations of the circuit, randomly assigning values to the components within their manufacturing tolerance limits. The resulting graph displays a tight band of overlapping waveforms. This distribution proves that, even with random variations, the output voltage remains highly stable and precisely clustered around the nominal value.

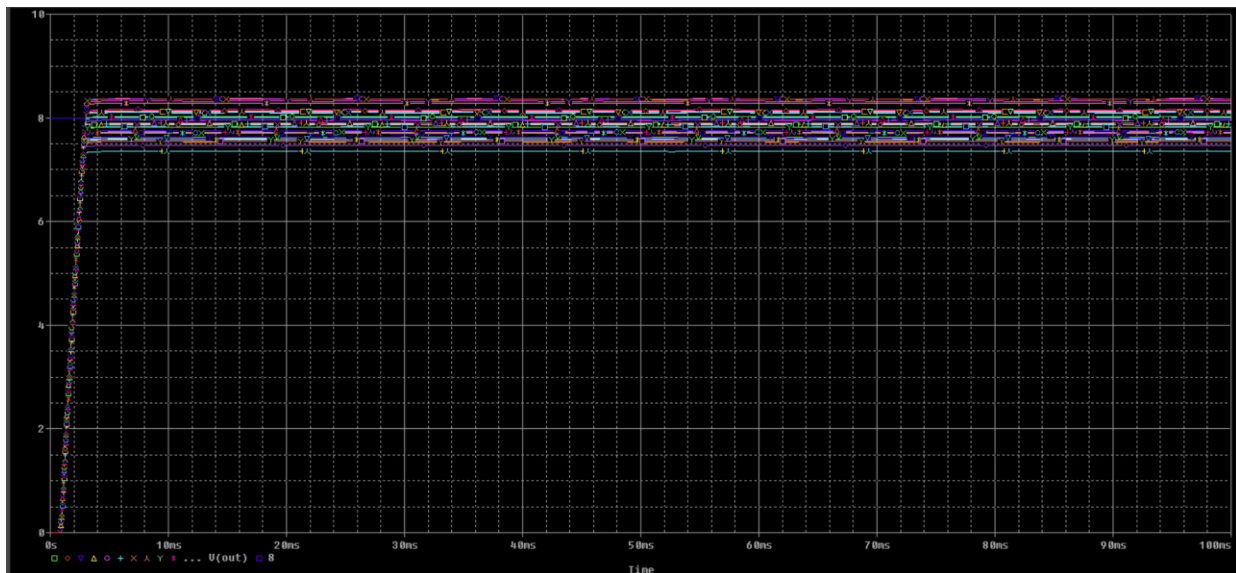


Figure 25 - Monte Carlo Tolerance Analysis

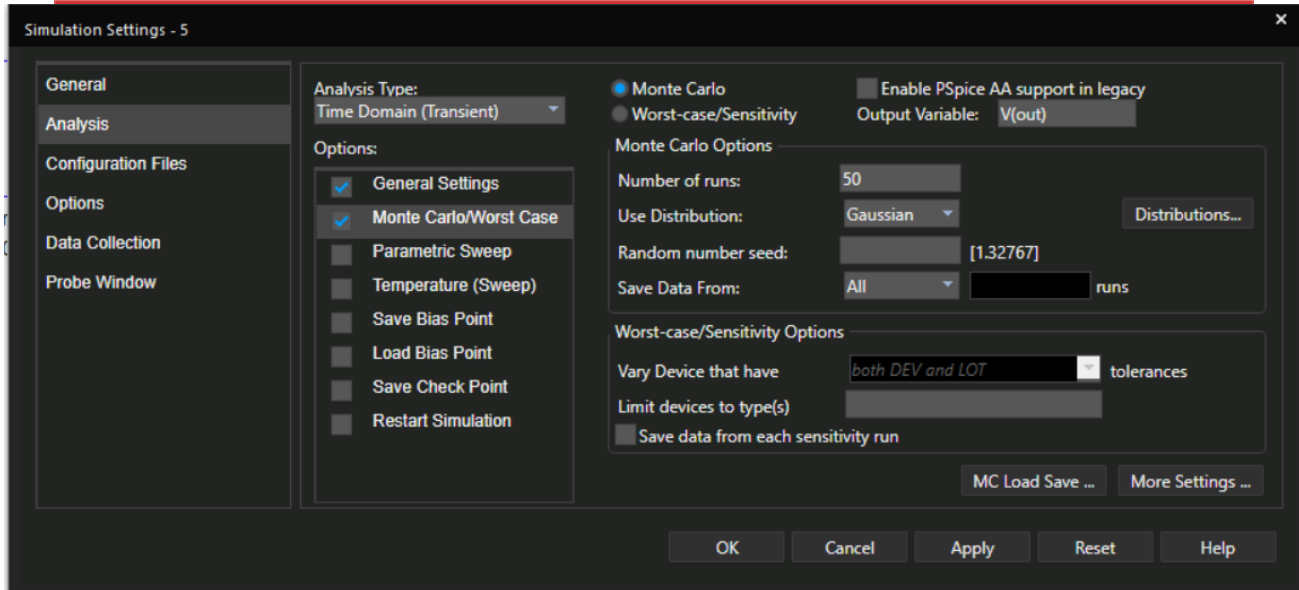


Figure 26 - Simulation Profile

## 5.5. Worst-Case Analysis

While the Monte Carlo analysis shows the average distribution, a Worst-Case analysis was performed to identify the absolute limits of the circuit. The simulation was configured to find the maximum possible deviation. The software intentionally pushed the tolerant components to their extreme limits (upper or lower), in the exact combination that would cause the largest voltage defect at the output. The results demonstrated that the maximum deviation is extremely small, proving that the design is safe and will not over-volt sensitive components even under the most unfavorable manufacturing conditions.

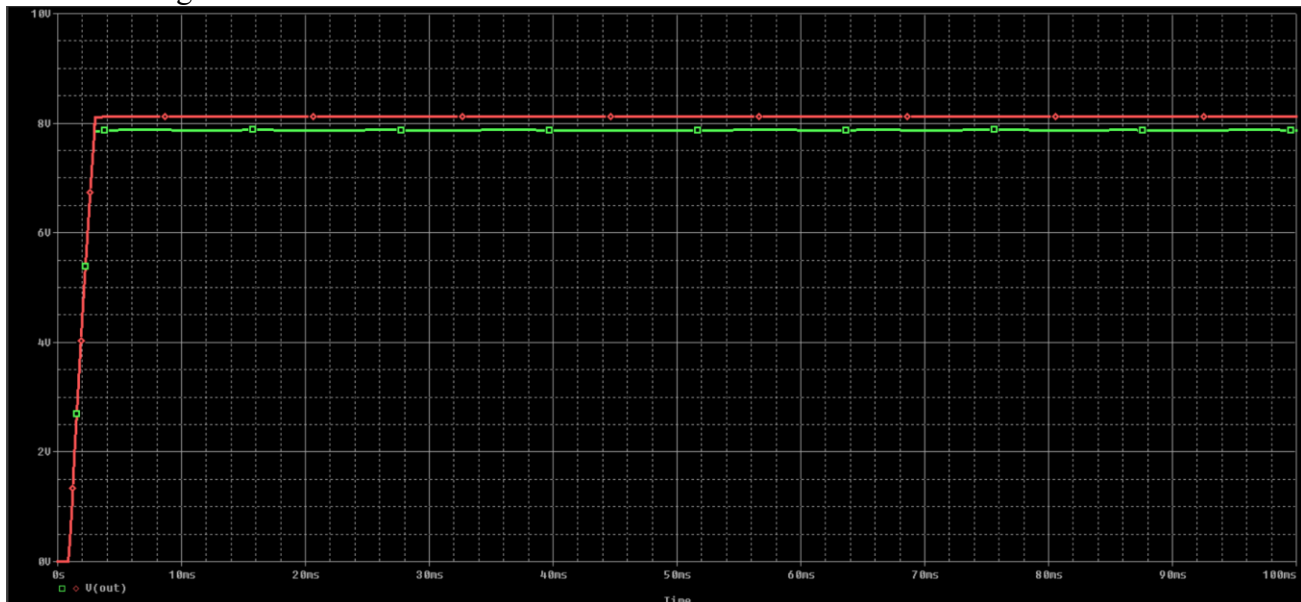


Figure 27 - Worst-Case Analysis

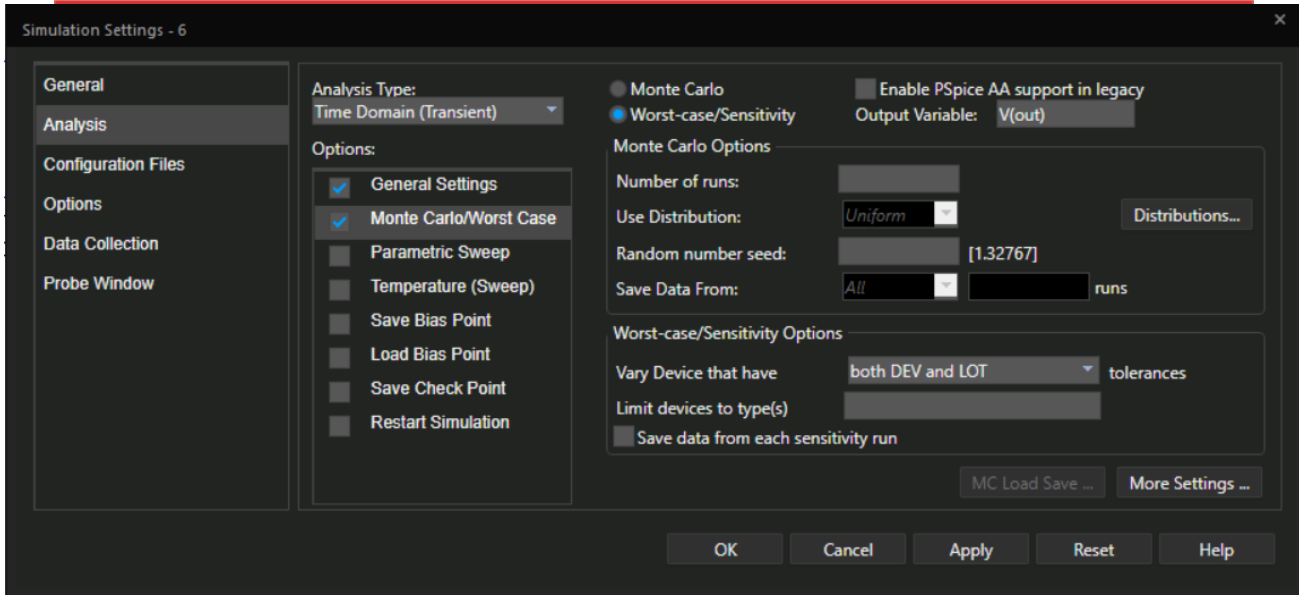


Figure 28 - Simulation Profile

This table details the specific component variations applied during the sensitivity analysis. To evaluate the absolute limits of the voltage regulator, the simulation software intentionally adjusted critical components, such as the feedback network and the Zener biasing resistor, to their extreme upper or lower 5 percent tolerance boundaries. The new value column indicates exactly which resistors were increased or decreased to forcefully create the most unfavorable operating conditions for the control loop.



Figure 29 - Sensitivity Summary

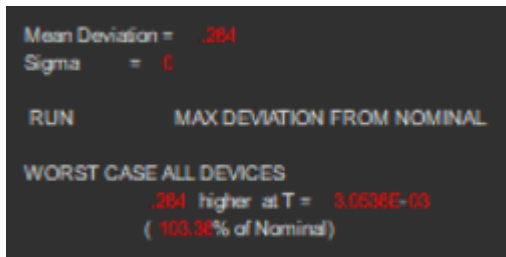


Figure 30 - Worst Case Summary

This summary presents the final numerical results of the worst-case tolerance test. It calculates the maximum possible deviation of the output voltage when all selected components simultaneously operate at their worst theoretical limits. As shown by the mean deviation metric, the difference between the ideal output voltage and the worst-case output is extremely small. This confirms that the operational amplifier and the voltage reference provide excellent stability, easily absorbing manufacturing imperfections without compromising the performance of the regulator.

## 6. Bill of materials

| Nr. Ord. | Component      | Comp           | Value         | Cant | Part code/description      | Price(\$) |
|----------|----------------|----------------|---------------|------|----------------------------|-----------|
| 1        | Transformer    | TX1            | 220V/15V (3A) | 1    | Hammond 185G16 (Step-down) | 14.50\$   |
| 2        | Diode          | D1, D2, D3, D4 | 3A, 400V      | 4    | 1N5404 (Power Rectifier)   | 0.60\$    |
| 3        | Capacitor      | C1, C2         | 3300uF        | 2    | Panasonic ECA-1EHG332      | 2.40\$    |
| 4        | Zener Diode    | D5             | 3.0V          | 1    | BZT55B3V0                  | 0.15\$    |
| 5        | Resistor       | Rd             | 500 ohms      | 1    | YAGEO CFR-25JR-52 (5%)     | 0.04\$    |
| 6        | OpAmp          | U1A            | -             | 1    | Texas Instruments LM358P   | 0.35\$    |
| 7        | Resistor       | R2             | 11.84 kOhms   | 1    | Vishay Precision (1%)      | 0.10\$    |
| 8        | Potentiometer  | Rpot           | 20 kOhms      | 1    | Bourns 3362P-1-203LF       | 0.65\$    |
| 9        | Transistor     | Q1, Q2         | NPN 15A       | 2    | STMicroelectronics 2N3055  | 3.20\$    |
| 10       | Transistor     | Q3             | NPN 0.8A      | 1    | 2N2222A (TO-92)            | 0.20\$    |
| 11       | Power Resistor | Rp             | 0.25 ohms     | 1    | YAGEO SQP500JB-0R25 (5W)   | 0.45\$    |
| 12       | Load Resistor  | RLOAD          | 10 ohms       | 1    | YAGEO SQP10A (Dummy Load)  | 0.55\$    |
|          |                |                |               |      | Total estimated price:     | 23.19\$   |

## 7. Conclusion

This project presents a robust linear voltage regulator designed to convert a standard 220V AC grid supply into a stable, adjustable DC output ranging from 3.0V to 8.0V. The system successfully utilizes a step-down transformer for galvanic isolation, a 1N5404 full-wave bridge rectifier, and a massive 6600 uF capacitive filter to deliver a clean DC baseline to the active regulation loop.

An LM358 operational amplifier, referenced against a highly stable 3.0V Zener diode, acts as the error amplifier. It effectively drives a Darlington pair of 2N3055 power transistors to maintain strict output stability regardless of load variations. Furthermore, an active overcurrent protection circuit, utilizing a 0.25 ohm sense resistor and a 2N2222 transistor, was implemented to safely and instantly clamp the output current at a maximum limit of 3.0A, protecting the hardware from short-circuit damage and thermal runaway.

Extensive PSpice testing, including time-domain Transient Analysis, Monte Carlo statistical evaluation, and Worst-Case tolerance checks, confirmed that the circuit operates flawlessly under nominal, heavy load, and extreme manufacturing conditions. Overall, this design provides a highly reliable, safe, and efficient solution for adjustable DC power supply requirements.

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